Republic of the Philippines

**ISABELA STATE UNIVERSITY**

Echague, Isabela — Main Campus

**COLLEGE OF COMPUTING STUDIES, INFORMATION AND COMMUNICATION TECHNOLOGY**

*Department of Computer Science • Data Mining Track*

# LETTER OF SYSTEM IMPLEMENTATION AND TURNOVER

| | |
|---|---|
| **DATE:** | November 12, 2026 |
| **TO:** | DR. FLORENDA B. TEMANEL — Executive Officer / Campus Administrator, ISU Echague Main Campus |
| **THRU:** | DR. ALBERT VINLUAN — Thesis Adviser / Research & Linkages Coordinator, CCSICT |
| **SUBJECT:** | **Formal System Implementation, Deployment, and Acceptance of "IskAI: A Centralized AI-Powered Thesis Library Using Retrieval-Augmented Generation"** |

**Dear Dr. Temanel:**

We are honored to respectfully endorse the complete deployment, technical turnover, and institutional implementation of the research software system titled **"A Centralized AI-Powered Thesis Library Using Retrieval-Augmented Generation" (IskAI)**, developed in partial fulfillment of the degree requirements for **Bachelor of Science in Computer Science (BSCS — Data Mining Track)** at the College of Computing Studies, Information and Communication Technology (CCSICT), Isabela State University – Echague Main Campus.

IskAI is deployed as an indirect, closed-domain academic thesis repository that eliminates the risks of ungrounded LLM hallucinations and protects raw manuscript intellectual property. Utilizing high-dimensional vector embeddings, hybrid semantic reranking, and LangChain orchestration with Google Gemini, the platform delivers verifiable, citation-backed answers strictly grounded in CCSICT's institutional research corpus.

**The deployed production environment comprises five operational subsystems:**

- **Closed-Domain RAG Pipeline:** LangChain + Supabase pgvector + Gemini 3.8 Flash with 800/100-token chunking and hybrid reranking for citation-stamped research assistance.
- **Proposal Novelty & Duplication Guard:** Screens Working Title, Rationale, Objectives, and Methodology against an 85% cosine similarity threshold to prevent duplicate research.
- **Academic Front-Matter Extraction:** Automatically parses Title, Authors, Year, Adviser, Panel, and Dual Abstracts (verbatim & AI-extended) upon ingestion.
- **Student Research Workspace:** Delivers interactive starter chips, multi-thesis comparative synthesis trays, and one-click APA 7th Edition / BibTeX citation copy.
- **Administrative Governance Console:** Enforces 5-year recency filtering, permanent archival retention, role-based security, and full operational audit logs.

All software deliverables, containerized production codebases, deployment configurations, database schemas, user manuals, and administrative credentials have been fully turned over. Comprehensive automated and user acceptance testing confirms that the system complies with ISO/IEC 25010 software quality standards.

**Respectfully submitted by the Proponents:**

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**AHRON JOHN F. BARLIS**
Lead Developer / BSCS Proponent
Date Signed: _____

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**CARLO ROSSI P. GALLARDO**
Co-Developer / BSCS Proponent
Date Signed: _____

**VERIFIED, ENDORSED, AND APPROVED BY:**

*Verified and Implemented by:*

_____

**JEN BERMUSA**
Technical Implementation Lead / IT Faculty
College of Computing Studies, Information & Comm. Tech.
Date Signed: _____

*Recommending Approval / Thesis Adviser:*

_____

**DR. ALBERT VINLUAN**
Thesis Adviser / Program Chair, BS Data Science
College Research & Linkages Coordinator, CCSICT
Date Signed: _____

**APPROVED AND ACCEPTED BY:**

_____

**DR. FLORENDA B. TEMANEL**
Executive Officer / Campus Administrator
Isabela State University — Echague Main Campus
Date Signed: _____

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# CERTIFICATE OF SYSTEM IMPLEMENTATION AND ACCEPTANCE

*Institutional Deployment and Technical Turnover Sign-Off*

**THIS IS TO CERTIFY that** the software system titled "A Centralized AI-Powered Thesis Library Using Retrieval-Augmented Generation" (IskAI), developed and submitted by AHRON JOHN F. BARLIS and CARLO ROSSI P. GALLARDO in partial fulfillment of the degree of Bachelor of Science in Computer Science (Data Mining Track), has been fully delivered, deployed, tested, and officially IMPLEMENTED and ACCEPTED for institutional operations at the College of Computing Studies, Information and Communication Technology (CCSICT).

## SYSTEM IMPLEMENTATION & OPERATIONAL ACCEPTANCE MATRIX:

| # | Subsystem / Deliverable | Scope of Operational Implementation | Status |
|---|---|---|---|
| 1 | **Closed-Domain RAG Retrieval Engine** | LangChain + Supabase pgvector + Gemini 3.8 Flash; 800/100-token overlap chunking, deterministic reranking, citation-backed answers. | **OPERATIONAL & ACCEPTED** |
| 2 | **4-Section Research Capsule Scanner** | Analyzes Title, Rationale, Objectives, Methodology; automated cosine similarity screening with 85% safety threshold. | **OPERATIONAL & ACCEPTED** |
| 3 | **Front-Matter Ingestion & Dual Abstract** | Automated extraction of Title, Authors, Advisers, Panelists; dual abstract mode (verbatim and AI-synthesized). | **OPERATIONAL & ACCEPTED** |
| 4 | **Student Workspace & Citation Tools** | Interactive query starter chips, 3-thesis comparative synthesis tray, APA 7th Edition and BibTeX bibliographic generation. | **OPERATIONAL & ACCEPTED** |
| 5 | **Administrative Governance Portal** | Role-based access control, 5-year recency filter toggle, inventory health checks, audit logs, and secure archive maintenance. | **OPERATIONAL & ACCEPTED** |
| 6 | **Source Code & Deployment Artifacts** | Complete Git repository, Docker containers, database migrations, technical deployment documentation, and admin credentials. | **DELIVERED & VERIFIED** |

*Issued this 12th day of November, 2026 at Isabela State University – Main Campus, Echague, Isabela, Philippines. This certification confirms that the system satisfies all operational, technical, and institutional requirements.*

## OFFICIAL SYSTEM ACCEPTANCE AND SIGN-OFF:

*Verified and Implemented by:*

_____

**JEN BERMUSA**
Technical Implementation Lead / IT Faculty
College of Computing Studies, Information & Comm. Tech.
Date Signed: _____

*Recommending Approval / Thesis Adviser:*

_____

**DR. ALBERT VINLUAN**
Thesis Adviser / Program Chair, BS Data Science
College Research & Linkages Coordinator, CCSICT
Date Signed: _____

**APPROVED AND ACCEPTED BY:**

_____

**DR. FLORENDA B. TEMANEL**
Executive Officer / Campus Administrator
Isabela State University — Echague Main Campus
Date Signed: _____